Parallel Tempering for Redistricting: Replica Exchange MCMC with Geometric Tolerance Ladders

B.20 — Algorithm Design / Search Layer

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Abstract

Single-chain Markov chain Monte Carlo (MCMC) redistricting algorithms—ReCom, ForestRecom, Merge-Split—can become trapped in local optima for large-k states such as Texas ($k = 38$) where the chain must pass through high edge-cut "barriers" to reach qualitatively different plan regions. We introduce ParallelTempering redistricting, a Search-layer algorithm that runs N_{replicas} chains simultaneously at different balance tolerances arranged on a geometric ladder: a cold chain at the statutory 0.5% tolerance and a hot chain at 5% tolerance, with intermediate chains in between. After every `swap_interval` steps, adjacent chains propose to exchange their current plans via a Metropolis criterion, $\min\big(1, \exp\big((\beta_i - \beta_j)(\mathrm{EC}_i - \mathrm{EC}_j)\big)\big)$, where $\beta = 1/\varepsilon$ is the inverse tolerance. Hot chains mix freely across plan space; the cold chain inherits this global diversity via accepted swaps.

Empirically, ParallelTempering with $N_{\text{replicas}} = 4$ achieves lower cold-chain edge cut than single-chain ForestRecom on Texas ($k = 38$), where local-optima trapping is most severe, with modest improvements on North Carolina ($k = 14$) and Wisconsin ($k = 8$) where single-chain mixing is adequate. Observed swap acceptance

rates of 18–24% are near the empirical target of ≈23% for Gaussian systems (Kone & Kofke 2005), providing evidence that the geometric ladder is well-calibrated for redistricting chains. The cold chain returns the distribution-correct plan; the hot chains serve only to inject global diversity. ParallelTempering is implemented as `-search parallel-tempering` in the `redist` binary and is the recommended search strategy for states with $k \geq 30$.

Contents

1. Introduction

1.1. The Local-Optimum Problem in Single-Chain MCMC

The B-series redistricting platform constructs congressional maps via a bisection tree: the state graph is recursively split into k population-balanced districts by minimising the weighted edge cut at each node [Della-Libera, 2026b]. The Search layer determines how many MCMC steps are taken and which resulting plan is returned. For most states with small to medium district counts ($k \leq 20$), a single ForestRecom chain run for $T = 1{,}000$ steps adequately explores the feasible plan space: the chain mixes freely within its balance-tolerance constraint and the minimum edge-cut plan is selected by percentile.

For large-k states, this single-chain strategy fails in a characteristic way. Texas ($k = 38$), California ($k = 52$), and Florida ($k = 28$) have state graphs with tens of thousands of census tracts. The feasible plan space at strict balance tolerance ($\varepsilon = 0.5\%$) has a rugged energy landscape: many local-optimum plans separated by high edge-cut "barriers." The single ForestRecom chain gets trapped in one basin of attraction; running for more steps explores the basin thoroughly but cannot escape to qualitatively different plan regions without passing through infeasible or high-EC intermediate states.

Track 1: Geographic coarsening. MultiScale (G.11) [Della-Libera, 2026d] addresses this by operating at a coarser spatial resolution: groups of tracts are merged into super-tracts, and the chain proposes swaps at the super-tract level. This enlarges the proposal neighbourhood and allows larger structural changes per step. The cost is that coarsening introduces approximation; the final plan is projected back to tract resolution.

Track 2: Temperature ladders. This paper introduces a second track: **Parallel Tempering** (PT), which runs N_{replicas} chains simultaneously, each at a different balance tolerance. Hot chains (high ε) mix freely because they accept almost any balanced plan; cold chains (low ε) produce distribution-correct samples but mix slowly. After every `swap_interval` steps, adjacent chains propose to exchange their current plans via a Metropolis criterion. Accepted swaps pass global diversity downward from the hot chains to the cold chain.

1.2. Parallel Tempering as a Redistricting Search Strategy

In statistical mechanics, parallel tempering (PT) is a method for sampling from complex energy landscapes [Swendsen and Wang, 1986, Geyer, 1991]. The "temperature" in statistical mechanics controls how strongly a system is constrained to low-energy states; high temperature permits wide exploration of state space. In redistricting, the analogous parameter is the population balance tolerance ε: a high tolerance accepts more plans as feasible, giving the chain a much larger proposal neighbourhood and faster mixing.

The key insight is that ε need not be fixed for all chains. In ParallelTempering redistricting:

- The *cold chain* ($\varepsilon_0 = 0.5\%$) targets the statutory distribution. Only its samples are used for plan selection.
- The *hot chain* ($\varepsilon_{N-1} = 5\%$) mixes rapidly because it can move between any roughly population-balanced plans. It serves only as a diversity source.
- Intermediate chains form a geometric ladder between cold and hot.

When a hot chain reaches a low-EC plan that is far from the cold chain's current location in plan space, a swap can transfer this plan to the cold chain, which then explores the new basin under tight balance constraints. This is the mechanism by which PT escapes local optima without modifying the distributional target of the cold chain.

1.3. Main Results

Algorithm. We define ParallelTempering redistricting (Section 3) as a Search-layer strategy that runs N_{replicas} ForestRecom chains on a geometric tolerance ladder with swap proposals at every `swap_interval` steps. The cold chain records all visited plans; the plan at the p-th percentile of the cold chain's edge-cut distribution is returned. The algorithm has four parameters: `n_replicas` (default 4), `swap_interval` (default 10), `cold_tol` (default 0.005), `hot_tol` (default 0.05).

Empirical. On TX ($k = 38$), ParallelTempering with $N = 4$ achieves lower cold-chain minimum edge cut than single-chain ForestRecom, demonstrating that replica exchange escapes the local optima where the single chain stalls. On NC ($k = 14$) and WI ($k = 8$), the improvement is modest—the single chain already mixes adequately for these states. Observed swap acceptance rates of 18–24% match the empirical target of ≈23% [Kone and Kofke, 2005], providing evidence that the geometric ladder is well-calibrated for redistricting chains.

Decision rule. ParallelTempering is the recommended search strategy when $k \geq 30$ and local-optimum trapping is suspected. For medium k ($8 \leq k \leq 20$), single-chain ForestRecom is preferred for its lower cost and equivalent quality. For large k with strong geographic structure, MultiScale [Della-Libera, 2026d] may provide better efficiency per replica.

1.4. Paper Structure

Section 2 reviews parallel tempering theory and the redistricting context. Section 3 gives the formal algorithm, pseudocode, seeding specification, and audit chain. Section 4 presents comparisons on NC, WI, and TX. Section 5 provides a structured decision guide. Section 6 situates ParallelTempering in the broader B/G-series algorithm landscape.

1.5. Relation to Prior Work

ParallelTempering is a Search-layer algorithm, orthogonal to the Structure-layer algorithms (GeoSection, AreaSection, SimulatedAnnealing) and the bisection-tree SeedCompositor strategies (ConvergenceSweep, BisectionEnsemble). Swendsen and Wang [1986] introduced replica exchange Monte Carlo for spin glasses; Geyer [1991] established the theoretical basis for using the cold chain output as a valid sample from the target distribution. Kone and Kofke [2005] derived the ≈23% optimal swap acceptance rate for Gaussian continuous systems; we adopt this as an empirical calibration target for discrete redistricting chains, not as a derived guarantee. MultiScale [Della-Libera, 2026d] is the closest B/G-series alternative, targeting the same large-k mixing problem via geographic coarsening rather than temperature ladders. ParallelTempering differs in mechanism: it requires no coarsening and preserves the exact ForestRecom proposal distribution at every tolerance level.

2. Background

2.1. Parallel Tempering in Statistical Mechanics

Parallel Tempering (PT) was introduced by Swendsen and Wang [1986] for simulating spin-glass systems at low temperature. At low temperature, a single Markov chain may become trapped in a local energy minimum for exponentially long times, exploring only a small fraction of the configuration space. Running multiple chains at increasing temperatures simultaneously allows high-temperature chains to mix rapidly and periodically propose their configurations as candidates for the low-temperature chain.

Geyer [1991] formalised PT as a statistical Monte Carlo method, establishing that the joint chain over all replicas is ergodic and that the marginal distribution of the cold replica is exactly the target distribution. This is the theoretical foundation for using only the cold chain's output in plan selection: the hot chains are sampling aids, not target distributions.

Temperature and mixing. In a physical system with energy $E(\sigma)$ and inverse temperature β, the Boltzmann acceptance criterion is $\min(1, \exp(-\beta\Delta E))$. High temperature (low β) means almost all proposals are accepted; the chain moves freely. Low temperature (high β) means only improvements or small increases are accepted; the chain explores a narrow basin of the energy landscape.

The replica exchange criterion between replica i (at β_i) and replica j (at $\beta_j > \beta_i$, so $\varepsilon_j > \varepsilon_i$) is:

$$P(\text{swap}) = \min\Big(1,\, \exp\Big((\beta_i - \beta_j)(E_i - E_j)\Big)\Big),$$

where $E_i = \text{EC}(\text{assignment}_i)$ is the edge cut of replica i's current plan. This is a valid Metropolis proposal on the joint distribution over all replicas, preserving the product-of-marginals target distribution [Geyer, 1991].

Balance tolerance as redistricting temperature. In redistricting, the population balance tolerance ε plays the role of temperature. A plan assignment $\mathbf{a} : V \to \{1, \ldots, k\}$ is *feasible at tolerance* ε if every district satisfies:

$$\left|\frac{\text{pop}(d)}{\bar{p}} - 1\right| \leq \varepsilon,$$

where $\bar{p} = P_{\text{total}}/k$ is the ideal population per district. At $\varepsilon = 0.005$ (statutory cold tolerance), only plans with ±0.5% population deviation are accessible. At $\varepsilon = 0.05$ (hot tolerance), plans with up to ±5% deviation are accessible—a much larger feasible set. The ForestRecom chain at high tolerance can make proposals that jump between topologically distinct plan types (e.g., moving a district's centre of gravity by several tracts in a single spanning-tree resampling), while the cold chain can only make local adjustments within the tight balance constraint.

The ≈23% swap acceptance target. Kone and Kofke [2005] showed that for Gaussian continuous distributions, the optimal swap acceptance rate between adjacent replicas is approximately 23%, achieved by spacing temperatures so that $\Delta(\ln T) \approx \text{const}$. For discrete combinatorial chains, this result is an empirical guideline rather than a derived theorem. Nevertheless, we use it as a calibration target for the geometric tolerance ladder: if observed swap acceptance rates are near 23% across all adjacent pairs, the ladder is well-spaced. Section 4 reports observed rates of 18–24% across NC, WI, and TX, consistent with this target.

2.2. ForestRecom as the Per-Replica Chain

Each replica in ParallelTempering redistricting runs one instance of ForestRecom [Della-Libera, 2026c], the spanning-forest recombination chain. ForestRecom proposes plan changes by:

1. Selecting two adjacent districts d_i, d_j.
2. Sampling a random spanning forest of the induced subgraph $G[d_i \cup d_j]$ with one component per district (a forest with exactly one edge cut between the two components).
3. Accepting the induced partition if the population balance constraint is satisfied at the replica's tolerance.

ForestRecom targets an approximately uniform distribution over balanced plans at its configured tolerance [Della-Libera, 2026c]. Each replica in ParallelTempering targets this uniform distribution at its own tolerance level; the cold chain's stationary distribution is the target we care about.

Forward and reverse RNG split. ForestRecom uses separate forward and reverse random number streams (for the forward proposal and the reverse-proposal computation in the Hastings ratio). In ParallelTempering, each replica's forward and reverse streams are derived from the replica seed via SHA-256 with distinct prefixes, as specified in Section 3.3.

2.3. Search-Layer Context in the B-Series Platform

The B-series redistricting platform uses a three-layer compositor: *Structure* (how the bisection tree is built), *WeightsOverride* (what edge weights signal), and *SeedCompositor* / *Search* (how the plan space is explored and the best plan selected). ParallelTempering is a Search-layer algorithm.

Existing search strategies. `single`: Run one ForestRecom chain for T steps, return the minimum-EC plan. `multi`: Run T independent chains from different seeds, return the minimum-EC plan overall. `convergence` (ConvergenceSweep, B.16): Run chains until $T = 600$ consecutive seeds produce no improvement. `percentile`: Run one chain, return the plan at the p-th percentile of the edge-cut distribution. `bisection-ensemble`: Run chains over a bisection-tree ensemble.

ParallelTempering is a new Search strategy: run N chains at different tolerances simultaneously, with periodic swap proposals, and return the plan at the p-th percentile of the *cold chain's* edge-cut distribution.

Orthogonality. ParallelTempering is orthogonal to the Structure and WeightsOverride layers. It can be combined with any structure configuration (GeoSection, ApportionRegions, standard bisect) and any weight configuration (geographic, county, unweighted). The empirical experiments in Section 4 use the GeoSection structure and geographic weights as the standard baseline.

2.4. Comparison with Multi-Scale ReCom

MultiScale [Della-Libera, 2026d] and ParallelTempering are both designed to improve mixing for large-k states. Their mechanisms differ:

Property	MultiScale (G.11)	ParallelTempering (B.20)
Mixing mechanism	Geographic coarsening	Temperature ladders
Proposal size	Large (super-tract level)	Standard (tract level)
Approximation	Yes (coarsening resolution)	No (exact ForestRecom at each ε)
Chain count	1	N_{replicas}
Extra cost	$\alpha\times$ coarsening overhead	$N_{\text{replicas}}\times$ per-step cost
Best for	Strong geographic structure	Tolerance-induced barriers

The two approaches are complementary and can in principle be combined (each PT replica running an MS chain at its tolerance), though this is left to future work. In practice, PT is preferred when the bottleneck is the balance-tolerance constraint rather than geographic coarsening, and MS is preferred when the bottleneck is proposal size.

3. The Parallel Tempering Algorithm

3.1. Notation and Setup

Let $G = (V, E, w)$ be the census-tract adjacency graph for a state, where V is the set of tracts, E is the set of shared-boundary edges, and $w : E \to \mathbb{R}_{>0}$ is the geographic boundary-length weight. Let $p : V \to \mathbb{Z}_{>0}$ be the 2020 Census tract populations, and let k be the target number of congressional districts. A *plan* is an assignment $\mathbf{a} : V \to \{1, \ldots, k\}$ such that every district $d = \mathbf{a}^{-1}(j)$ is connected in G and satisfies the population balance constraint at tolerance ε:

$$\left| \frac{\sum_{v\in d} p(v)}{P/k} - 1 \right| \le \varepsilon,$$

where $P = \sum_{v\in V} p(v)$ is the state total population. The *edge cut* of a plan is

$$\mathrm{EC}(\mathbf{a}) = \sum_{\{u,v\}\in E:\ \mathbf{a}(u)\neq\mathbf{a}(v)} w(u, v).$$

3.2. Geometric Tolerance Ladder

ParallelTempering runs $N = N_{\text{replicas}}$ chains indexed $i = 0, 1, \ldots, N-1$. Replica i operates at balance tolerance ε_i drawn from a *geometric ladder*:

$$\varepsilon_i = \varepsilon_{\text{cold}} \times \left(\frac{\varepsilon_{\text{hot}}}{\varepsilon_{\text{cold}}}\right)^{i/(N-1)}, \quad i = 0, 1, \ldots, N-1.$$

This gives $\varepsilon_0 = \varepsilon_{\text{cold}}$ (the cold chain, strict statutory tolerance) and $\varepsilon_{N-1} = \varepsilon_{\text{hot}}$ (the hot chain, relaxed tolerance). All intermediate tolerances are strictly increasing.

Definition 1 (Inverse tolerance / inverse temperature). *For replica i with tolerance ε_i, the* inverse tolerance *(the analogue of inverse temperature) is $\beta_i = 1/\varepsilon_i$. The cold chain has the highest β (most constrained); the hot chain has the lowest β (least constrained).*

Equal swap acceptance design. The geometric ladder is chosen so that the expected swap acceptance rate between adjacent replicas is approximately equal across all pairs. For continuous Gaussian systems, equal spacing in $\ln \varepsilon$ achieves $\approx$23% swap acceptance [Kone and Kofke, 2005]. For discrete redistricting chains, this serves as an empirical starting point. The ladder spacing can be diagnosed from the observed swap acceptance rates in the audit log; a rate substantially below 15% suggests over-spacing (too many replicas needed), and a rate substantially above 35% suggests under-spacing (fewer replicas would suffice).

Default parameters.

Parameter	Default	Notes
N_{replicas}	4	4 chains at $\varepsilon \in \{0.005, 0.013, 0.033, 0.05\}$
$\varepsilon_{\text{cold}}$	0.005	±0.5% statutory tolerance
ε_{hot}	0.05	±5% relaxed tolerance
`swap_interval`	10	Swap proposals every 10 steps
T (steps)	1000	Total cold chain steps
p (percentile)	0.0	Select minimum-EC cold chain plan

3.3. Seeding Specification

All random streams are derived from a single `base_seed` via SHA-256 with domain-separated prefixes, following the platform-wide seed design. Four distinct prefixes are used:

Definition 2 (Replica Seed)**.**

$$\texttt{replica\_seed}(b, r, s) = \text{first8}\Big(\text{SHA-256}(\textit{"PT\_REPLICA\_"} \parallel \text{le4}(r) \parallel \textit{"\_"} \parallel \text{le8}(s) \parallel \textit{"\_"} \parallel \text{le8}(b))\Big),$$

where b is the base seed (u64), r is the replica index (u32), s is the step index (u64), le4 *and* le8 *encode values as little-endian bytes, and* first8 *reads the first 8 bytes of the SHA-256 digest as a little-endian u64.*

Definition 3 (Swap Seed)**.**

$$\texttt{swap\_seed}(b, s, i) = \text{first8}\Big(\text{SHA-256}(\textit{"PT\_SWAP\_"} \parallel \text{le4}(i) \parallel \textit{"\_"} \parallel \text{le8}(s) \parallel \textit{"\_"} \parallel \text{le8}(b))\Big),$$

where i is the pair index (0-based, for the swap between replicas i and $i+1$).

The forward and reverse RNG streams for replica r at step s are derived from the replica seed:

$$\texttt{rng\_fwd} = \text{SmallRng} :: \text{seed\_from\_u64}\Big(\text{first8}(\text{SHA-256}(\texttt{"PT\_FWD\_"} \parallel \text{le8}(r_s)))\Big),$$

$$\texttt{rng\_rev} = \text{SmallRng} :: \text{seed\_from\_u64}\Big(\text{first8}(\text{SHA-256}(\texttt{"PT\_REV\_"} \parallel \text{le8}(r_s)))\Big),$$

where $r_s = \texttt{replica\_seed}(b, r, s)$.

Remark 1. *The four prefixes "PT_REPLICA_", "PT_SWAP_", "PT_FWD_", and "PT_REV_" are all distinct. In particular,* $\texttt{replica\_seed}(b, r, s) \neq \texttt{swap\_seed}(b, s, r)$ *for all (b, r, s) because the prefixes differ even when the numeric arguments coincide. A test asserts that each prefix constant in source equals its expected string and that* $\texttt{replica\_seed}(0, 0, 0)$ *and* $\texttt{swap\_seed}(0, 0, 0)$ *equal hard-coded expected values, preventing silent seed-compatibility drift.*

3.4. Formal Pseudocode

Algorithm 1 gives the complete ParallelTempering procedure.

Three design choices. **1. Geometric tolerance ladder.** Equal spacing in $\ln \varepsilon$ gives approximately equal swap acceptance rates across all adjacent pairs. A ladder with equal absolute spacing $\Delta\varepsilon$ would concentrate swaps near the cold end (where the tolerance difference matters most) and produce near-universal acceptance near the hot end. The geometric ladder distributes swap events more uniformly.

2. Swap interval $= 10$**.** The swap interval balances two competing objectives. If swaps are proposed too frequently (interval $= 1$), the chains do not have time to diffuse between swap events; consecutive swaps on the same pair are anti-correlated, reducing effective exploration. If swaps are proposed too infrequently (interval $= 100$), the hot chain may return to the cold chain's basin before the swap is proposed, losing the diversity benefit. Interval $= 10$ is the standard practical choice from the tempering literature [Earl and Deem, 2005] and is confirmed by the swap acceptance rates in Section 4.

3. Cold chain recording. The cold chain records all plans it visits, not only the plans where an internal ForestRecom proposal was accepted. Regardless of whether the ForestRecom proposal was accepted, the cold chain is in some plan (either the proposed plan or the previous one); we record that plan. This means $|\mathcal{R}| = T + 1$ (the initial plan plus one record per step). The percentile selection at line 25 is over all cold chain states, including states that persisted across multiple steps due to proposal rejection.

Cold chain acceptance rate vs. swap acceptance rate. Two acceptance rates are tracked and reported in the audit log:

- `swap_acceptance_rate`: Fraction of swap proposals that were accepted across all adjacent pairs. Measures chain exchange health.
- `cold_chain_acceptance_rate`: Fraction of ForestRecom steps at replica 0 where the internal proposal was accepted. Measures cold chain health.

A very low cold chain acceptance rate with a normal swap rate indicates that the cold tolerance is too tight for this state: the cold chain cannot move at $\varepsilon_{\text{cold}}$, but the PT mechanism has no way to help because the cold chain rejects proposals before swaps can inject diversity. A normal cold chain acceptance rate with very low swap rate indicates that the tolerance ladder is over-spaced: adjacent chains are too different in tolerance to exchange efficiently.

3.5. CLI Invocation and YAML Configuration

ParallelTempering is exposed via the `-search parallel-tempering` flag:

```
redist state --state TX --year 2020 \
  --search parallel-tempering \
  --pt-replicas 4 \
  --pt-swap-interval 10 \
  --pt-cold-tol 0.005 \
  --pt-hot-tol 0.05 \
  --seeds 1000 \
  --percentile 0.0
```

YAML configuration:

```
algorithm:
  structure: ratio-optimal
  weights: geographic
  search: parallel-tempering
  pt_replicas: 4
  pt_swap_interval: 10
  pt_cold_tolerance: 0.005
  pt_hot_tolerance: 0.05
  percentile: 0.0
```

3.6. Audit Chain

Every run appends the following fields to `runs/{label}/{year}/index.json`:

```
"search": "parallel-tempering",
"n_replicas": 4,
"swap_interval": 10,
"cold_tolerance": 0.005,
"hot_tolerance": 0.05,
"percentile": 0.0,
"base_seed": 12345678,
```

```
"steps": 1000,
"swap_acceptance_rate": 0.21,
"cold_chain_acceptance_rate": 0.44,
"selected_step": 847,
"replica_seed_formula": "SHA-256('PT_REPLICA_' || r:u32le || '_' || s:u64le || '_' || b:
"swap_seed_formula":    "SHA-256('PT_SWAP_'    || i:u32le || '_' || s:u64le || '_' || b:
```

The field `selected_step` is the *rank* of the returned plan in the cold chain's EC-sorted record list (0 = minimum EC), not a temporal step index. An auditor who re-runs the chain with the same `base_seed` collects all cold chain records, sorts by (EC ASC, step ASC), and confirms that the plan at rank `selected_step` matches the submitted plan.

Legal provenance of parallel chains. A special master or opposing expert may ask: does using N replicas change the distributional properties of the submitted plan compared to a single-chain ForestRecom run? The answer is no: the submitted plan is drawn from the cold chain ($\varepsilon = \varepsilon_{\text{cold}}$, the same tolerance as single-chain ForestRecom), and the cold chain's stationary distribution is the same ForestRecom target. The hot replicas serve as exploration aids; they do not affect the cold chain's targeting. The audit chain records all N chains' seeds, swap events, and the cold-chain `selected_step`, enabling full independent verification.

3.7. Computational Cost

ParallelTempering costs $N_{\text{replicas}} \times T$ ForestRecom steps plus $\lfloor T/\texttt{swap\_interval} \rfloor \times (N_{\text{replicas}} - 1)$ swap evaluations. Each swap evaluation requires computing EC for two chain states, which costs $O(|E|)$ per plan. In practice, EC can be maintained incrementally across ForestRecom steps so that the per-swap cost is $O(1)$ (only the boundary edges of the changed district pairs need updating). The dominant cost is the $N_{\text{replicas}} \times T$ ForestRecom steps, which is N times the cost of a single-chain run. For $N = 4$, the wall-clock cost is approximately $4\times$ the baseline single-chain run time (assuming serial chain execution); with parallel execution via Rayon, the wall-clock cost approaches the baseline (open question, noted in Section 6).

4. Empirical Results

4.1. Experimental Setup

We compare three algorithms on three states chosen to span the range of congressional delegation sizes and geographic complexity:

- **NC** (North Carolina, $k = 14$, $n \approx 2{,}660$ tracts): medium complexity, competitive two-party state.
- **WI** (Wisconsin, $k = 8$, $n \approx 1{,}900$ tracts): small delegation, historically contested.
- **TX** (Texas, $k = 38$, $n \approx 5{,}300$ tracts): large delegation, geographically diverse with major metropolitan areas. This is the primary target state where single-chain local-optimum trapping is most pronounced.

All runs use 2020 Census data, geographic boundary-length edge weights (B.2 default [Della-Libera, 2026a]), the GeoSection ratio-optimal tree structure (B.8), and $T = 1{,}000$ steps per chain. The three algorithms compared are:

1. **ForestRecom ($N = 1$)**: Single ForestRecom chain, $T = 1{,}000$ steps, $\varepsilon = 0.005$. This is the standard baseline.
2. **PT ($N = 2$)**: ParallelTempering with 2 replicas, cold $\varepsilon = 0.005$, hot $\varepsilon = 0.05$, `swap_interval`$= 10$.
3. **PT ($N = 4$)**: ParallelTempering with 4 replicas, cold $\varepsilon = 0.005$, hot $\varepsilon = 0.05$, `swap_interval`$= 10$. Intermediate tolerances: $\varepsilon_1 \approx 0.013$, $\varepsilon_2 \approx 0.033$.

We also report comparison figures for MultiScale [Della-Libera, 2026d] on TX, where available from G.11.

Reproducibility note. All values in Table 1 are single-run estimates (base seed $s = 42$). We verified directional robustness across seeds $s \in \{42, 43, 44, 45, 46\}$: the EC ordering $\mathrm{PT}(N = 4) \leq \mathrm{PT}(N = 2) \leq$ ForestRecom single-chain holds for all five seeds on TX, with run-to-run EC variation $< 4\%$.

Metrics. For each (state, algorithm) pair we report:

- **Min EC**: Minimum edge cut observed in the cold chain's record list (equivalently, the plan returned at percentile $p = 0$).
- **Mean EC**: Mean edge cut over all cold chain records.
- **Swap Rate**: Observed swap acceptance rate across all adjacent replica pairs and all swap opportunities.
- **Cold Accept**: Cold chain internal ForestRecom acceptance rate.
- **Runtime ratio**: Wall-clock time relative to single-chain ForestRecom ($N = 1$), on a single core.

4.2. Results

Table 1 presents the comparison. ParallelTempering($N = 4$) achieves lower minimum edge cut than single-chain ForestRecom in all three states, with the largest improvement on Texas ($k = 38$) where local-optimum trapping is most severe. The swap acceptance rates of 18–24% are consistent with the ≈23% empirical target from Kone and Kofke [2005], providing evidence that the geometric ladder is well-calibrated for redistricting chains.

4.3. Analysis of Results

Large-k benefit. The EC improvement from ForestRecom($N = 1$) to PT($N = 4$) is 3.3% for NC, 3.4% for WI, and 7.6% for TX. The substantially larger improvement on TX confirms the local-optimum trapping hypothesis: the single chain on TX gets stuck in a basin with EC ≈ 11,440 from which it cannot escape without the diversity injection provided by hot chain swaps. After an accepted swap, the cold chain finds itself in a different plan region and can explore from there; its internal ForestRecom proposals (at $\varepsilon = 0.5\%$) now find lower-EC plans that were inaccessible from the original basin.

Geographic interpretation of TX local optima. The 7.6% EC reduction on TX corresponds to a qualitative geographic difference: the single-chain ForestRecom plan tends to keep the Houston metropolitan area as a single compact block, while the PT($N = 4$) plan sometimes produces an alternative partition where Harris County is split differently

Table 1: Algorithm comparison on NC ($k = 14$), WI ($k = 8$), TX ($k = 38$). Min EC: minimum edge cut over cold chain record list (lower is better). Mean EC: mean edge cut over cold chain (lower is better). Swap Rate: observed swap acceptance rate. Cold Accept: cold chain internal acceptance rate. Runtime: wall-clock time relative to single-chain ForestRecom ($N = 1$).

State	Algorithm	Min EC	Mean EC	Swap Rate	Cold Accept	Runtime
	ForestRecom ($N = 1$)	4,821	4,953	—	0.47	1.0×
NC ($k = 14$)	PT ($N = 2$)	4,716	4,891	0.21	0.44	2.0×
	PT ($N = 4$)	4,663	4,841	0.22	0.43	4.0×
	ForestRecom ($N = 1$)	2,103	2,174	—	0.51	1.0×
WI ($k = 8$)	PT ($N = 2$)	2,059	2,138	0.23	0.50	2.0×
	PT ($N = 4$)	2,031	2,110	0.23	0.49	4.0×
	ForestRecom ($N = 1$)	11,440	11,892	—	0.38	1.0×
TX ($k = 38$)	PT ($N = 2$)	10,893	11,421	0.19	0.36	2.0×
	PT ($N = 4$)	10,571	11,198	0.18	0.35	4.0×

between districts. Both configurations are geographically contiguous and VRA-compliant; they represent distinct local optima in the compactness landscape. This illustrates that PT's value is not merely numerical — it finds plans that single-chain methods would not reach within practical runtimes.

Autocorrelation and mixing quality. To assess whether the EC improvement reflects improved mixing or favourable seeds, we compute the lag-100 autocorrelation of the cold-chain EC trace. For TX $k = 38$: PT($N = 4$) ACF(100) = 0.31 vs. single-chain ForestRecom ACF(100) = 0.53 — a 42% reduction consistent with the escape-from-local-optima mechanism. For NC $k = 14$, ACF values are comparable (0.38 vs. 0.41), consistent with NC already mixing adequately with a single chain.

Replica count scaling. Adding replicas beyond $N = 2$ provides diminishing but positive returns. PT($N = 2$) already captures most of the benefit: the step from $N = 1$ to $N = 2$ accounts for approximately 60–70% of the total PT($N = 4$) improvement. The step from $N = 2$ to $N = 4$ adds a further 30–40%. This is consistent with the empirical observation that a two-replica system already achieves substantial mixing improvement when the ladder is geometric; adding replicas further refines the coverage but with diminishing marginal returns [Earl and Deem, 2005].

Swap acceptance rates. The observed swap rates of 18–24% are near the ≈23% empirical target. The TX chains show slightly lower swap rates (18–19%) because the larger state has more edge-cut variance across replicas—each replica's current EC is drawn from a wider distribution, which reduces the probability that a cold-hot swap is accepted. This is consistent with the Kone-Kofke theory: wider EC distributions between replicas correspond to a less ideal geometric spacing. For TX with $N = 6$ replicas (as recommended in the parameter scaling table of the spec), the swap rates are expected to return to the 21–24% range.

Cold chain acceptance rates. The cold chain acceptance rates (0.35–0.51) are healthy across all states. The lowest rate is TX (0.35 for PT($N = 4$)): the cold chain at $\varepsilon = 0.5\%$ has a somewhat lower ForestRecom acceptance rate on Texas than on NC or WI, because the balance constraint is harder to satisfy in a large state with concentrated metropolitan populations. This rate is not pathologically low; a rate below 0.10 would indicate that the cold tolerance is too tight for the state's population distribution.

Comparison with Multi-Scale. For TX, MultiScale [Della-Libera, 2026d] achieves a minimum EC of approximately 10,380 (from G.11), compared to PT($N = 4$)'s 10,571. MultiScale has an edge on TX because the geographic coarsening allows larger structural changes per step—it effectively operates at a coarser spatial resolution that spans the metropolitan–suburban boundary, which is the primary local-optimum barrier on Texas maps. PT is competitive with MultiScale and does not require a coarsening heuristic; for states where geographic coarsening is difficult to calibrate (complex coastlines, archipelago counties), PT may be preferred.

NC and WI: modest but positive improvement. For NC ($k = 14$) and WI ($k = 8$), the single-chain ForestRecom already mixes adequately and PT provides a modest but positive improvement of 3–4%. This is expected: when the chain mixes well without hot chain assistance, the replica exchange mechanism still occasionally injects useful diversity from the hot chain's broader sampling of plan space. The cost is $N\times$ the runtime; for NC and WI, the runtime ratio of $4\times$ for PT($N = 4$) is a significant overhead for a modest gain.

For these states, single-chain ForestRecom or ConvergenceSweep (convergence sweep) is the recommended search strategy.

5. When to Use Parallel Tempering

5.1. Decision Guide

ParallelTempering is the right tool when the single-chain MCMC is stuck. The following conditions identify states where local-optimum trapping is likely:

Condition 1: Large k. States with $k \geq 30$ (TX, CA, FL, NY) have complex, high-dimensional plan spaces where the ForestRecom chain at $\varepsilon = 0.5\%$ cannot easily traverse the energy barriers between distinct plan regions. The feasible set at $\varepsilon = 0.5\%$ is a small fraction of the feasible set at $\varepsilon = 5\%$; the hot chain can traverse barriers in the looser set and occasionally hand configurations to the cold chain via swaps.

Condition 2: Known local optima. If repeated runs from different seeds consistently produce the same EC value (or a small cluster of values), the chain is likely trapped. This is detectable from the audit log: if `cold_chain_ec_variance` across multiple runs is small relative to the range of EC values seen in the hot chain, the cold chain is trapped.

Condition 3: Willing to pay the replica cost. PT costs $N_{\text{replicas}}\times$ the single-chain runtime (serial execution). If runtime is a binding constraint (e.g., interactive use or rapid iteration), PT is not appropriate. For publication-quality maps or statutory submissions, the runtime cost is usually acceptable.

5.2. Algorithm Decision Matrix

Situation	Recommended	Reason	CL
$k \leq 5$, fast iteration	`single`	Adequate mixing, minimal cost	`-se`
$k = 6$–20, single best plan	`convergence`	CS sweeps seeds until convergence	`-se`
$k = 6$–20, distributional coverage	`percentile`	Full cold chain distribution	`-se`
$k \geq 30$, best plan needed	**PT** ($N = 4$)	Replica exchange escapes local optima	`-se`
$k \geq 30$, geographic structure strong	MultiScale	Coarsening matches geographic barriers	`-se`
$k \geq 30$, auditable + reproducible	**PT** ($N = 4$)	Exact seed derivation, full audit chain	`-se`
VRA constraints present	VraRecom	VRA-aware chain preserves minority districts	`-se`

5.3. PT vs. ForestRecom vs. Multi-Scale

The three strategies for large-k states are single-chain ForestRecom, PT, and MultiScale. Their trade-offs are:

ForestRecom (baseline).

- *Strength*: Cheapest, simplest, exact distributional target at cold ε.
- *Weakness*: Stalls on TX-class states; no mechanism to escape local optima.
- *When*: Always run as baseline for comparison. Use as final algorithm only for $k \leq 20$.

PT.

- *Strength*: No approximation. Hot chains explore broadly; cold chain inherits diversity. Exact ForestRecom proposal distribution at every tolerance level. Strong audit chain (all seeds deterministic from `base_seed`).
- *Weakness*: $N\times$ runtime cost (serial). Hot chains do not contribute directly to the output—they are overhead.

- *When*: $k \geq 30$, local optima suspected, runtime acceptable, strong auditability required. Default for Texas, California, Florida, New York.

MultiScale.

- *Strength*: Larger proposals—can move entire clusters of tracts simultaneously. Better efficiency per replica for states where the local-optimum barrier corresponds to a geographic boundary (e.g., urban/suburban split in TX).
- *Weakness*: Coarsening introduces approximation. Requires calibrating the coarsening resolution α for each state. Complex coastlines or non-contiguous counties may cause coarsening failures.
- *When*: $k \geq 30$, state has strong geographic structure (river systems, mountain ranges) that defines the local-optimum barriers, and coarsening can be calibrated reliably.

Combining PT and MultiScale. In principle, each PT replica could run an MultiScale chain at its tolerance, combining the global diversity injection of PT with the large-proposal benefit of MultiScale. This is not implemented in the current platform and is left as an open question for G-series future work.

5.4. Parameter Selection for Large States

For large-k states ($k \geq 30$), the recommended PT configuration is:

- $N_{\text{replicas}} = 6$: Additional replicas improve ladder coverage. The step from $N = 4$ to $N = 6$ is more valuable for large states (where $N = 4$ swap rates drop to 18%, below the optimal target) than for small states.
- `cold_tol` $= 0.005$: Statutory tolerance, fixed.
- `hot_tol` $= 0.10$: Looser than the default 0.05 for large states, where the energy barrier is higher and a wider hot-chain sampling range is needed to bridge it.
- `swap_interval` $= 20$: Longer between swaps for large states, giving each chain more time to diffuse between swap events.

These recommendations follow from the parameter scaling table in the spec (Section 1 of the specification document) and are supported by the empirical swap acceptance rates in Section 4.

6. Conclusion

We have introduced ParallelTempering redistricting, a Search-layer algorithm that runs N_{replicas} ForestRecom chains simultaneously at different balance tolerances arranged on a geometric ladder. Adjacent chains periodically propose to exchange their current plans via the Metropolis criterion; accepted swaps inject global diversity from hot chains into the cold chain, enabling escape from local optima without modifying the cold chain's distributional target.

Key design choices. Three design choices make ParallelTempering redistricting practical:

1. The *geometric ladder* $\varepsilon_i = \varepsilon_c \times (\varepsilon_h/\varepsilon_c)^{i/(N-1)}$ provides approximately equal swap acceptance rates across all adjacent pairs, calibrated by the ≈23% empirical target from continuous systems [Kone and Kofke, 2005].

2. The *swap interval* of 10 steps balances chain diffusion time against swap frequency, consistent with established tempering practice [Earl and Deem, 2005].

3. The *cold chain record collection* stores all $T + 1$ cold chain states, sorts by EC, and returns the plan at the requested percentile. This allows percentile-based plan selection without rerunning the chain.

Empirical summary. PT($N = 4$) achieves 7.6% EC reduction over single-chain ForestRecom on Texas ($k = 38$), 3.4% on Wisconsin ($k = 8$), and 3.3% on North Carolina ($k = 14$). The improvement is largest where local-optimum trapping is most severe. Observed swap acceptance rates of 18–24% confirm that the geometric ladder is well-calibrated across all three states.

Position in the algorithm landscape. ParallelTempering is a Search-layer algorithm, orthogonal to the Structure layer (GeoSection, ApportionRegions, SimulatedAnnealing) and to the WeightsOverride layer. It composes with any structure/weights configuration. The recommended decision rule for practitioners is:

- $k \leq 20$: Use single-chain ForestRecom or ConvergenceSweep. PT overhead not justified for adequate mixing states.
- $k \geq 30$**, auditability priority**: Use PT ($N = 4$ default, $N = 6$ for TX/CA). The SHA-256 seed derivation provides a complete audit chain from `base_seed` to the selected plan.
- $k \geq 30$**, geographic structure strong**: Consider MultiScale or PT+MultiScale (future work).

Implementation. ParallelTempering redistricting is implemented in the `redist` binary under `-search parallel-tempering`, with parameters `-pt-replicas`, `-pt-swap-interval`, `-pt-cold-tol`, and `-pt-hot-tol`. The `ParallelTemperingChain` struct lives in `redist-ensemble/src/pa` and exposes `swap_acceptance_rate()`, `cold_chain_ec_mean()`, and `select_plan(p)` as the primary API.

Open questions.

1. **Parallel execution.** The N chains at different tolerances are independent between swap steps—an embarrassingly parallel workload. Rayon-based parallel execution would reduce wall-clock cost from $N\times$ to approximately $1\times$ the single-chain runtime, making PT competitive on wall-clock time even for large N. This is deferred as a Phase 2 implementation task.
2. **Mixed-chain design.** Should hot chains use standard ReCom (faster per step, higher acceptance) while only the cold chain uses ForestRecom (distributional correctness)? Mixed-chain designs complicate the swap Metropolis criterion but may improve wall-clock performance.
3. **Adaptive replica count.** Adding or removing replicas during the run based on observed swap acceptance rates would optimise the ladder dynamically. This requires

careful audit-chain handling to maintain reproducibility.

4. **PT × Multi-Scale.** Each PT replica could run an MultiScale chain at its tolerance. This would combine the global diversity injection of PT with the large-proposal benefit of MultiScale, potentially providing the best of both mechanisms for TX-class states.

5. **Non-adjacent swaps.** Current implementation only proposes swaps between adjacent pairs $(i, i + 1)$. Non-adjacent swaps (between any pair of replicas) may improve global mixing but require adjusting the Metropolis criterion and are less standard in the tempering literature.

These extensions inherit the clean three-layer compositor design of the existing platform and require no structural changes to the bisection tree or the ForestRecom chain implementation.

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Algorithm 1 $\textsc{ParallelTempering}(G,\ k,\ b,\ N,\ \varepsilon_c,\ \varepsilon_h,\ \texttt{swap\_interval},\ T,\ p)$

▷ Geometric tolerance ladder
for $i = 0$ **to** $N-1$ **do**
  $\varepsilon_i \leftarrow \varepsilon_c \times (\varepsilon_h/\varepsilon_c)^{i/(N-1)}$
end for
▷ Initialise all replicas from the same METIS plan
$\mathbf{a}_{\text{init}} \leftarrow \textsc{Metis}\text{-}k\text{-plan}(G, k, \varepsilon_c, b)$
for $i = 0$ **to** $N-1$ **do**
  $\texttt{chain}_i \leftarrow \textsc{ForestRecom}\text{-Chain}(G, k, \mathbf{a}_{\text{init}}, \varepsilon_i)$
end for
$\mathcal{R} \leftarrow \Big[(\text{EC}(\mathbf{a}_{\text{init}}),\ 0,\ \mathbf{a}_{\text{init}}.\text{clone}())\Big]$ ▷ cold chain records: (EC, step, assignment)
for $s = 1$ **to** T **do**
  ▷ Step all replicas independently
  for $i = 0$ **to** $N-1$ **do**
    $r_s \leftarrow \texttt{replica\_seed}(b, i, s)$
    $\texttt{rng\_fwd} \leftarrow \text{SmallRng}(\text{SHA-256}(\texttt{"PT\_FWD\_"} \| r_s))$
    $\texttt{rng\_rev} \leftarrow \text{SmallRng}(\text{SHA-256}(\texttt{"PT\_REV\_"} \| r_s))$
    $\texttt{chain}_i.\text{step}(\texttt{rng\_fwd},\ \texttt{rng\_rev})$
  end for
  $\mathcal{R}.\text{push}\Big((\text{EC}(\texttt{chain}_0.\mathbf{a}),\ s,\ \texttt{chain}_0.\mathbf{a}.\text{clone}())\Big)$
  ▷ Swap proposals every swap_interval steps
  if $s \bmod \texttt{swap\_interval} = 0$ **then**
    for $i = 0$ **to** $N-2$ **do**
      $\texttt{rng}_{\text{swap}} \leftarrow \text{SmallRng}(\texttt{swap\_seed}(b, s, i))$
      $\text{EC}_i \leftarrow \text{EC}(\texttt{chain}_i.\mathbf{a})$
      $\text{EC}_j \leftarrow \text{EC}(\texttt{chain}_{i+1}.\mathbf{a})$
      $\beta_i \leftarrow 1/\varepsilon_i; \quad \beta_j \leftarrow 1/\varepsilon_{i+1}$
      $\ell \leftarrow (\beta_i - \beta_j) \times (\text{EC}_i - \text{EC}_j)$
      if $\texttt{rng}_{\text{swap}}.\text{gen} ::< \text{f64} >().\ln() < \min(0, \ell)$ **then**
        swap $\texttt{chain}_i.\mathbf{a} \leftrightarrow \texttt{chain}_{i+1}.\mathbf{a}$
      end if
    end for
  end if
end for
sort $\mathcal{R}$ by (EC ASC, s ASC)
return $\mathcal{R}[\lfloor p \times |\mathcal{R}| \rfloor].\mathbf{a}$